

# Real-time multi-objective optimization of pump scheduling in water distribution networks using neuro-evolution

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## ABSTRACT

Pump scheduling in water distribution networks (WDNs) influences energy efficiency and supply reliability. Recently, machine-learning technologies showed promise in real-time control. However, methods solely focused on minimizing operational costs often limit decision-makers choices. This study introduces a real-time multi-objective optimization method for pump scheduling in WDNs employing neuro-evolution, with neural networks trained by NSGA-II. The approach explores neural network-based control policies to balance system resilience and operational cost, comparing their performance against two baseline methods using NSGA-II, i.e., scenario-specific optimization (SSO) and robust optimization (RO). Simulation results of the Anytown network show that neuro-evolution performs between SSO and RO in Pareto front hypervolume, with improved outcomes using a smaller-scale neural network and a larger population. Although neuro-evolution is inferior to RO in Pareto front sparsity, it performs better than SSO and RO in pipe failure scenarios. Under default water demand scenarios, over 99 % neuro-evolution policies effectively prevent water pressure deficiencies, surpassing SSO's 83 %. Across all testing scenarios, the least-cost neuro-evolution control policy shows a 2.7 % reduction in mean operational cost and achieves an approximately an 8 % higher minimum water supply ratio compared to RO. Neuro-evolution shows promise for multi-objective real-time scheduling but needs improved performance in Pareto front sparsity.

## 1. Introduction

A water distribution network (WDN) serves as a complex and critical urban infrastructure system, providing water security for both residents and businesses. Operational requirements (e.g. pump scheduling) in WDNs have garnered significant attention because of considerable energy consumption [1]. Typically, the operational goal of water utilities is to deliver a reliable and safe water supply to customers in the most cost-effective manner [2]. Despite the application of various optimization methods for pump operations [3–5], they often involve constrained single-objective optimization formulations. Consequently, solutions prioritizing minimal operational costs may restrict decision-making flexibility [6], particularly when considering potential system failures like pipe bursts. In practice, achieving a comprehensive understanding of the trade-offs between operational cost and system performance is preferable in the absence of a unique optimal solution.

Recently, machine-learning-based techniques, such as deep reinforcement learning (DRL), have shown potential in real-time control

problems in the water industry [7,8]. Unlike heuristic algorithms, such as genetic algorithms, employed for online optimization, DRL can learn control policies in advance, thus avoiding the need for online optimization during application [9]. There have been several attempts to apply DRL to real-time pump scheduling in WDNs [10–12], demonstrating its computational efficiency for real-time control. However, these efforts have not yet systematically explored the relationships between operational cost and system performance within a multi-objective framework. Further exploration of these relationships is necessary to support real-time decision-making in practice.

Rule-based control can also realize real-time control through pre-established rules [13], but it often relies on expert knowledge [14]. As the system complexity increases, developing rule-based solutions that reliably perform under all conditions becomes increasingly challenging. In contrast, DRL uses deep neural networks to approximate functions, offering a potential solution to the curse of dimensionality associated with the large number of possible system states and actions [15].

Neuro-evolution serves as an alternative to DRL for addressing

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decision-making problems [16], utilizing evolutionary algorithms to train neural networks [17]. Some studies have demonstrated that neuro-evolution can achieve comparable performance to DRL in domains such as Atari games [18] and humanoid locomotion [19]. However, the application of neuro-evolution in real-time control of urban water systems remains relatively unexplored, especially in terms of tackling complex system complexity, dynamics and uncertainties.

Resilience is a well-studied concept for evaluating infrastructure system performance [20–22]. It can measure a WDN's ability to handle unexpected events, such as pipe bursts. While resilience has been extensively studied in the design of WDNs [23–26], its application in operational optimization has received comparatively less attention. Although Odan et al. [2] and Zhang et al. [6] incorporated a resilience index for multi-objective operation optimization in WDNs, the heuristic algorithms employed for online optimization in these studies may impose a heavy computational burden during operation [27], thereby limiting their applicability for real-time control.

The aim of this study is to employ a neuro-evolution approach for real-time multi-objective optimization of pump scheduling in a WDN (i.e., the Anytown network). The optimization model incorporates a resilience index and operational cost simultaneously. A set of neural network-based control policies are trained using neuro-evolution and subsequently compared to baseline methods using NSGA-II across various water demand scenarios of the Anytown network. Two primary aspects are investigated in this study: (1) The performance comparison between the trained control policies and the baseline methods using Pareto front approximation based on the two objectives; (2) The performance evaluation of the trained control policies under pipe failures.

## 2. Case study

The Anytown network serves as the case study here, which has been modified based on the least cost design solution presented in Farmani et al. [28]. There are 19 demand nodes, 3 tanks, 41 pipes and 3 pumps in the Anytown network. The hydraulic simulation of the Anytown network is based on the simulator, EPANET.

### 2.1. Water demand scenario generation

To evaluate the performance of the pump scheduling methods considering the variations in actual demand, water demand scenarios are randomly generated using Latin hypercube sampling and a truncated normal distribution:

- (1) A random number is sampled for each node using Latin hypercube sampling from a truncated normal distribution  $N(1, 0.5)$ . In this study, the upper and lower bounds of the distribution are 0.7 and 1.3 respectively.
- (2) At each time step in the scheduling period, the hourly demand at each node is generated by multiplying its base demand by the relevant demand multiplier and by the random number calculated in Step (1).
- (3) The two steps above are repeated for each time step in the scheduling period to generate one demand scenario. For the Anytown network, there are a total of 150 demand scenarios generated: 100 used for training and 50 for testing. The scheduling period of each demand scenario is 24 h. Furthermore, for each water demand scenario, the hydraulic simulation starts at 6:00 PM, and the initial levels of all tanks are set to 10 m.

### 2.2. Problem formulation

#### 2.2.1. Objective functions and decision variables

For each water demand scenario of the Anytown network, there are two operational objectives for pump scheduling. The first objective is minimization of total operational cost in the scheduling period, given as

follows:

$$\text{Min} \sum_t \left( u_t \sum_{m=1}^3 E_{m,t} \right) \quad (1)$$

where  $u_t$  is the electricity tariff (i.e., 0.12 \$/kwh).  $E_{m,t}$  denotes the energy consumption of pump  $m$  at timestep  $t$ .

The second objective is maximizing the system resilience. The modified resilience index (MRI) proposed by Jayaram and Srinivasan [23] is applied here to assess system resilience. Unlike other resilience indicators for WDN analysis, such as the Resilience Index [26], the Network Resilience Index [25] and the Available Power Index [24], MRI is distinguished by its positive correlation between the surplus nodal head and system resilience. This characteristic makes MRI more suitable for system resilience assessment in pump scheduling problems. Consequently, the corresponding objective function is presented below:

$$\begin{aligned} &\text{Max} \frac{1}{k} \sum_{t=1}^k \text{MRI} \\ &s.t. \text{MRI} = \frac{\sum_n Q_n^{req} (H_n - H_n^{req})}{\sum_n Q_n^{req} H_n^{req}} \end{aligned} \quad (2)$$

where  $k$  is the number of decision steps in the scheduling period.  $N$  is the number of demand nodes.  $H_n$ ,  $H_n^{req}$  and  $Q_n^{req}$  are the actual head, the required head and the required flow rate at the demand node  $n$ . As depicted in Eq. (2), the MRI is defined as the ratio of the surplus energy for demand nodes to their required energy. Consequently, a higher mean value of MRI caused by the pump schedules indicates greater energy provided to the demand nodes generally. This can be interpreted as a favorable outcome in terms of system resilience.

In this study, the pump scheduling problem is to determine the time-schedule for pump switches during the scheduling period. The Anytown network includes three identical pumps, with a scheduling timestep of 1 h. To represent the state of all pumps at each timestep, the decision variable takes discrete values from 0 to 3. Specifically:

- 0 represents all pumps off
- 1 represents one pump on
- 2 represents two pumps on
- 3 represents all three pumps on

This encoding scheme was chosen because the pumps are identical. Using a single variable to capture the number of activated pumps at each timestep simplifies the representation of pump statuses and efficiently encapsulates the scheduling decisions.

#### 2.2.2. Constraints

There are two constraints set up for pump scheduling.

- (1) Node pressure requirement. To ensure sufficient water to users in the scheduling period, the nodal pressure at each timestep should be no less than the required minimum pressure, as below:

$$H_{nt} \geq H^{req} \quad (3)$$

where  $H_{nt}$  is the actual pressure of node  $H_n$  at timestep  $t$ .  $H^{req}$  is the required minimum pressure for each demand node (i.e., 40 psi).

- (2) Pump switch. Frequent pump switches can contribute to wear-out failures, consequently increasing the maintenance cost [29]. Therefore, the number of pump switches should be constrained to be lower than the maximum allowable times for a given scheduling period.

$$NS_m \leq NS_{max} \quad (4)$$

where  $NS_m$  is the actual number of switches for pump  $m$ .  $NS_{max}$  is the maximum allowable number of switches, which is set to 4 for each pump in the 24-hour pump scheduling.

### 3. Methodology

#### 3.1. Multi-objective optimization methods

##### 3.1.1. Neuro-evolution

In this study, neuro-evolution is employed for real-time multi-objective pump scheduling. Neuro-evolution realizes real-time pump scheduling by training the control policies represented by the neural networks (NNs) in advance, thus avoiding the need for online optimization process in the testing period. To characterize the trade-offs between operational cost and system resilience, a set of NN-based control policies are required. A widely-used multi-objective algorithm, NSGA-II [30] is used here for neuro-evolution.

The workflow of control policy training is shown in Fig. 1. Defining the NN structure, a population of NNs are initialized with different hyperparameters (i.e.,  $\theta_1, \theta_2, \dots, \theta_P$ ), representing different control policies (i.e.,  $\pi_{\theta_1}, \pi_{\theta_2}, \dots, \pi_{\theta_P}$ ). The hyperparameters include weights and bias.  $P$  is the population size. The control policies are evaluated by interacting with the training scenarios of the Anytown network. At timestep  $t$ ,  $\pi_{\theta_i}$  outputs the control actions  $a_{t+1}$  according to the given state information  $s_t$  as inputs, i.e.,  $a_{t+1} = \pi_{\theta_i}(s_t)$ . The state information comprises the levels of the three tanks, while the actions correspond to the number of operating pumps. The control policies are updated by NSGA-II iteratively based on the evaluation results of the training scenarios (see Eqs. (1)–(4)), including mean MRI, mean operational cost and constraint violations. The goal of the control policy training is to optimize a set of non-dominated control policies for the two operational objectives.

The NSGA-II implementation relies on the python Package, Platypus (<https://platypus.readthedocs.io/>). In this study, the number of generations for NSGA-II is 100. However, the number of NN hyperparameters to be optimized may affect the performance of the trained control policies. Hence, four two-hidden-layer NN structures are defined respectively to investigate this impact. The only distinction between these structures is the number of neurons in each hidden layer, respectively set at 5, 10, 30 and 50. Accordingly, the corresponding number of NN hyperparameters are 74, 194, 1174, and 2954, respectively. The trained neural network-based control policies are denoted as “NE5,” “NE10,” “NE30,” and “NE50” to signify the number of neurons in each hidden layer. Additionally, the tanh function is employed as the activation function for each hidden layer, and the population size is set at 100 for NSGA-II.

Furthermore, to investigate the impact of population size, the neural network with 30 hidden neurons is specifically employed. Three population sizes—100, 200, and 500—are set for NSGA-II to train control policies, denoted as “NEp100,” “NEp200,” and “NEp500” respectively.

In the testing stage, the set of the trained control policies are applied to each testing demand scenario respectively for performance evaluation. Based on the actions (i.e., pump solutions) generated by the trained control policies, the trade-off for each demand scenario is characterized. However, it is possible that the pump solutions of certain control policies are dominated by others (See Fig. 2). In this regard, the non-dominated pump solutions will be used for tradeoff characterization while the dominant pump solutions will be excluded. In addition, the number of points in Fig. 2 for a testing demand scenario may be less than the population size defined in NSGA-II, indicating instances where the pump solutions output by some control policies are identical.

##### 3.1.2. Baseline methods

To evaluate the performance of the trained control policies, two baseline methods based on NSGA-II are used here.

- (1) Scenario-specific optimization (SSO). SSO is an on-line optimization method that optimizes a specific set of pump solutions for each testing demand scenario according to the objectives and the constraints formulated above (see Eqs. (1)–(4)). NSGA-II is implemented with 200 generations and a population size of 500. SSO leverages comprehensive hydraulic information and extensive trial-and-error evaluation, allowing it to achieve optimal performance under known and stable conditions. However, SSO's

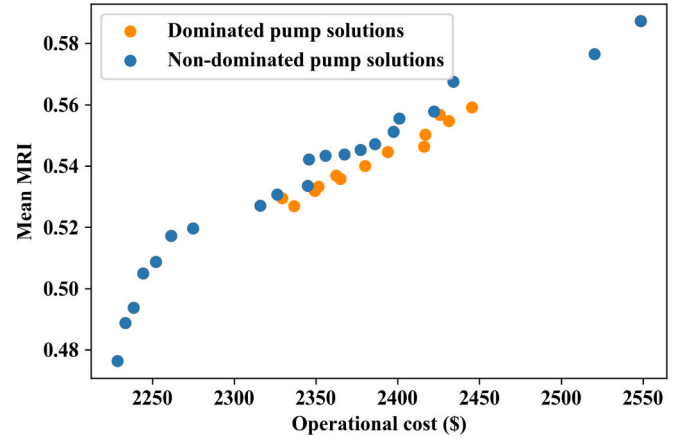


Fig. 2. Schematic diagram of the trade-off characterization using neuro-evolution.

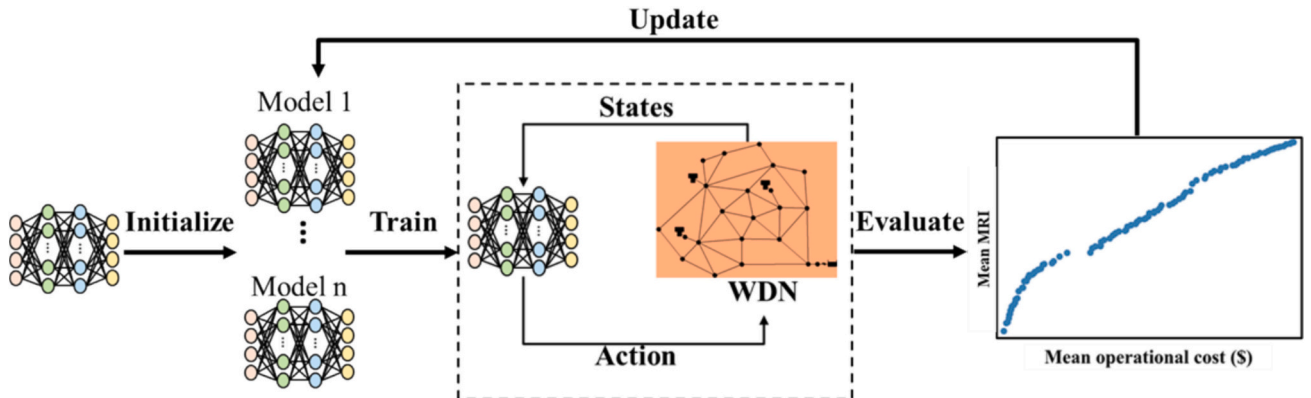


Fig. 1. NN models training for multi-objective real-time control in water distribution networks.

practical application is limited by its reliance on comprehensive hydraulic data and high computational costs, making it less suitable for real-time control scenarios. Consequently, SSO's optimization outcomes can be viewed as the theoretical upper bound for the performance of neuro-evolution under deterministic conditions.

- (2) Robust optimization (RO). RO is implemented for the search of a universal set of pump solutions considering the demand variations of the training scenarios. In other words, for the 100 training scenarios, RO aims to optimize pump solutions to maximize overall resilience (based on Eq. (2)) and minimize overall operational cost (based on Eq. (1)) without any constraint violations (based on Eqs. (3)–(4)). The set of the optimized pump solutions can then be directly applied to the testing scenarios, making RO a form of real-time scheduling. NSGA-II is implemented with 200 generations and a population size of 100 for RO.

### 3.2. Performance evaluation

#### 3.2.1. Pareto front approximation

For each testing demand scenario, the pump solutions in the trade-off are non-dominated, constituting what is referred to as the Pareto set here. The goal of multi-objective optimization methods is to approximate optimal Pareto set of each scenario. To measure the quality of the approximated Pareto fronts based on different methods, two metrics are used in this study.

- (1) Hypervolume metric [31], which quantifies the space dominated by the Pareto set in the objective space. As shown in Fig. 3, the hypervolume metric of an approximated Pareto front is the shaded area dominated by the approximated Pareto front and dominating the reference point. Hence, the definition of the reference point affects the hypervolume metric.

To determine this metric, an effective reference point is required for each demand scenario. Specifically, for a given demand scenario, the reference point (X, Y) is set as follows: X is defined as 101 % of the maximum observed operational cost across the non-dominated pump solutions evaluated under SSO, RO, and neuro-evolution control policies, while Y is defined as 99 % of the minimum MRI among those solutions.

Furthermore, to evaluate the comparative performance in term of hypervolume metric, a hypervolume ratio is introduced. This ratio for neuro-evolution is calculated by comparing its Pareto front hyper-

volume metric to that of SSO. A higher hypervolume ratio denotes superior performance of neuro-evolution, while a lower ratio indicates the opposite. Similarly, the hypervolume ratio of RO is calculated by comparing to SSO. The overall hypervolume ratio distributions of neuro-evolution and RO are obtained by collecting the hypervolume ratios of all testing scenarios separately.

- (2) Sparsity metric [32], which measures the sparsity of the Pareto set. In the two-objective optimization problems, the sparsity metric is calculated as the average distance between consecutive points in the approximated Pareto front. In the case of Fig. 3, the sparsity is calculated as  $\frac{1}{3}(d_0^2 + d_1^2 + d_2^2)$ .

A Pareto front of higher quality is expected to exhibit a higher hypervolume metric and a lower sparsity metric.

#### 3.2.2. Performance evaluation under pipe failures

As the resilience index, MRI, is a surrogate measure, the performance of the trained control policies in terms of system resilience need to be further evaluated under failure scenarios. A WDN using resilient pump solutions is expected to continue providing sufficient water to different demand nodes even during failure scenarios.

In this study, various pipe failures resulting from pipe burst are simulated. As shown in Fig. 4(a), the three water mains of the Anytown network are individually selected for failure simulation because of their significance in water supply. Since a pipe failure can occur at any time during the scheduling period, 22 possible start times are selected for the failure simulations (see Fig. 4(b)). Each simulated failure lasts for 3 h. Meanwhile, for simplification, the time required for pipe failure detection is neglected [33] and only the isolation required for pipe repair is simulated under the failure events. Thus, a pipe failure event is modeled by setting the status of the specific pipe to be closed for 3 h from the specified starting time [6]. In this context, 66 pipe failures are simulated for a testing demand scenario, resulting in a total of 3300 pipe failures for across the 50 testing demand scenarios.

To evaluate the performance of pump solutions under these pipe failures, two indicators are used:

- (1) Number of pressure deficient events (NP). A pipe failure event may cause the pressure at certain demand nodes to fall below the required level, violating the pressure requirement constraint. These instances are termed pressure deficient events. The number of pressure deficient events is counted based on the simulations of all pipe failure events.
- (2) Minimum ratio of water supply (RWS) during pressure deficient events. Water supply service can be adversely affected during any pressure deficient event. The RWS in the pipe failure duration is calculated for each pressure deficient event by comparing the actual water supply to the required amount. The minimum RWS is determined by identifying the smallest RWS among all pressure deficient events. To accurately simulate the actual water amount for each demand node, pressure driven analysis is utilized in the demand model for hydraulic simulations, as opposed to demand driven analysis (see [https://epanet22.readthedocs.io/en/latest/back\\_matter.html#input-file-format](https://epanet22.readthedocs.io/en/latest/back_matter.html#input-file-format) for more details).

## 4. Results and discussion

### 4.1. Performance of the Pareto front approximation

#### 4.1.1. Overall evaluation

The performance of neuro-evolution and the baseline methods for the Pareto front approximation is analyzed. This evaluation involves two key metrics for each demand scenario: hypervolume and sparsity.

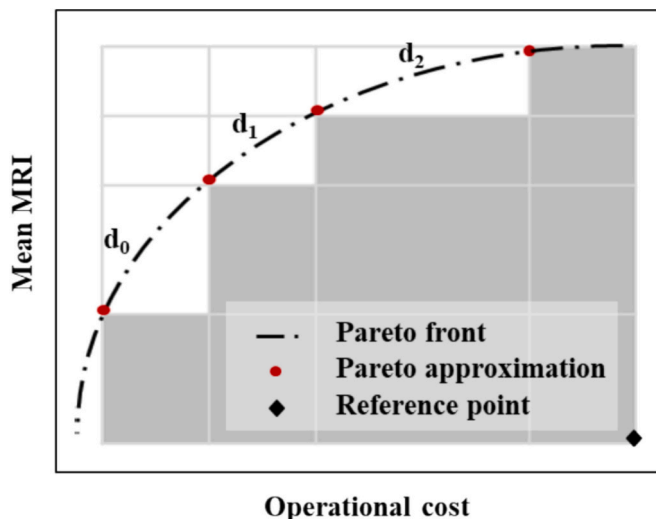


Fig. 3. Schematic diagram for the evaluation of the approximated Pareto front.

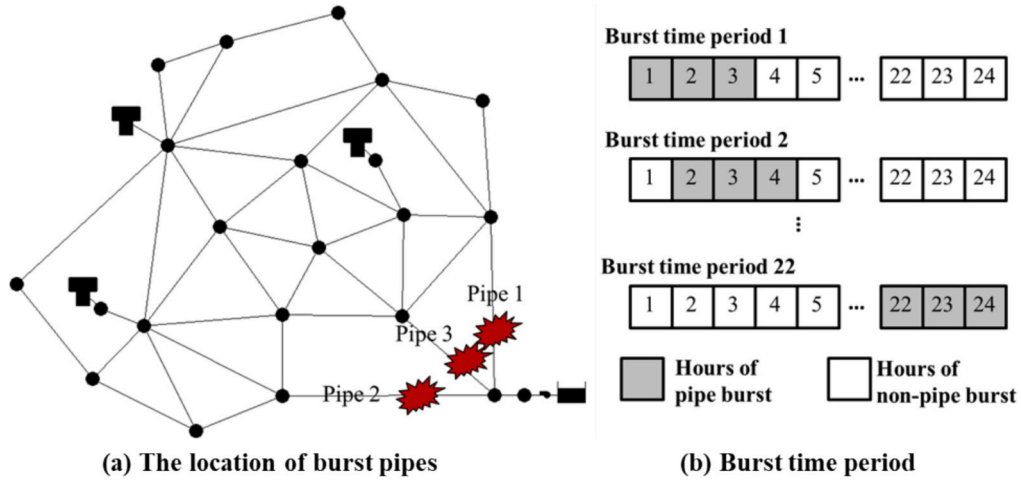


Fig. 4. Simulation of pipe failures.

Fig. 5(a) and (b) illustrates the hypervolume ratio distributions for the approximated Pareto fronts of neuro-evolution, with median values denoted by blue numbers in the boxplots. Fig. 5(a) shows a general decrease in the hypervolume ratio as the number of NN hidden neuron increases. This trend suggests that adding more neurons requires optimizing additional NN parameters, which negatively impacts the training process of neuro-evolution control policies. Conversely, a larger population size positively affects the hypervolume ratios, as depicted in Fig. 5(b). Comparing the hypervolume ratios of RO with those of various neuro-evolution control policies, it reveals that “NE5”, “NE10”, “NEp200” and “NEp500” have higher median hypervolume ratios than “RO”. This implies that, after training, neuro-evolution can outperform RO. In addition, the hypervolume ratio variations across different neuro-evolution control policies are comparatively smaller than the results of RO, indicating that the trained neuro-evolution control policies are more robust to demand uncertainty. However, all the hypervolume ratios of neuro-evolution control policies are below 1, suggesting that SSO has a better performance.

The sparsity distributions for the approximated Pareto fronts of neuro-evolution are shown in Fig. 5(c) and (d). In Fig. 5(c), the performance of neuro-evolution becomes more robust to demand uncertainty with an increase in the NN hidden neuron number, although the median value of the boxplot remains stable. The results in Fig. 5(d) show that the Pareto front sparsity of “NEp500” is lower than that of “NEp100” and “NEp200”, suggesting that a larger population size may contribute to a denser Pareto front. However, RO outperforms neuro-evolution in this respect, as its sparsity distribution is significantly lower than that of various neuro-evolution control policies. One factor contributing to this is that RO includes more non-dominated pump solutions in the approximated Pareto front compared to neuro-evolution (see Fig. 5(e) and (f)).

Moreover, all non-dominated solutions based on the neuro-evolution control policies and the baseline methods are evaluated for constraint violations across various testing demand scenarios. Among the 50 testing demand scenarios, all non-dominated pump solutions of SSO and RO satisfy the node pressure constraint and the pump-switch constraint (see Eqs. (3) and (4)). The total number of constraint violations via the neuro-evolution control policies is shown in Table 1. For each set of control policies, the proportion violating the node pressure constraint or the pump switch constraint is much less than 1 %. Thus, SSO, RO and neuro-evolution all perform well in preventing constraint violations.

The simulation results of the Anytown network reveal that neuro-evolution’s performance in terms of the Pareto front hypervolume metric falls between SSO and RO. Utilizing a smaller-scale neural network and a larger population size in the NSGA-II algorithm improves the

hypervolume of the approximated Pareto front. However, neuro-evolution exhibits inferior performance in the approximated Pareto fronts sparsity compared to RO. Therefore, further exploration is needed to enhance the quality of the approximated Pareto front through neuro-evolution.

#### 4.1.2. Detailed analysis of the default water demand scenario

The default water demand scenario of the Anytown network is used for further analysis to compare the performance of SSO, RO and neuro-evolution. The base node demand distribution for this scenario is shown in Fig. 6(a), while the approximated Pareto fronts of SSO, RO and neuro-evolution are shown in Fig. 6(b) and (c). As depicted, SSO entirely dominates all neuro-evolution control policies, except for the pump solution with the maximum MRI. This is because SSO, being an online optimization method, provides well-performing non-dominated solutions tailored to specific water demand scenarios. The gap between the approximated Pareto fronts of SSO and neuro-evolution is relatively small in the high resilience area but increases as resilience decreases. Comparatively, neuro-evolution’s approximated Pareto front covers a broader resilience range in contrast to RO. While RO dominates in the high resilience area, it struggles to provide effective pump solutions in the low resilience (or low operational cost) area. Additionally, RO’s narrower resilience range of the approximated Pareto front also contributes to its lower sparsity compared to neuro-evolution.

The two endpoints of the approximated Pareto fronts of SSO, RO and NEp500 are selected for detailed analysis, as depicted in Fig. 6(c). The pump scheduling solutions related to points A, B, C and D are illustrated in Fig. 7(a) with resilience variations in the scheduling period presented in Fig. 7(b). Point A, positioned at the intersection of all Pareto fronts, where all pumps are operational at each scheduling step, results in the highest operational cost as well as mean MRI. Point C represents the least-operational-cost pump solutions of RO, where all pumps are open at most scheduling steps. Consequently, Point C results in higher mean MRI and operational cost when comparing to that of SSO and neuro-evolution (i.e., points B and D).

#### 4.2. Performance evaluation under pipe failures

In this study, the incorporation of the surrogate indicator of WDN resilience, the MRI, into the optimization model aims to enhance solutions or control policies for handling unexpected events. This section evaluates the performance of SSO, RO and neuro-evolution under pipe failure events using four indicators: NP, minimum RWS in the pressure deficient events, mean operational cost of all pipe failure events and number of pump-switch constraint violations (NV).

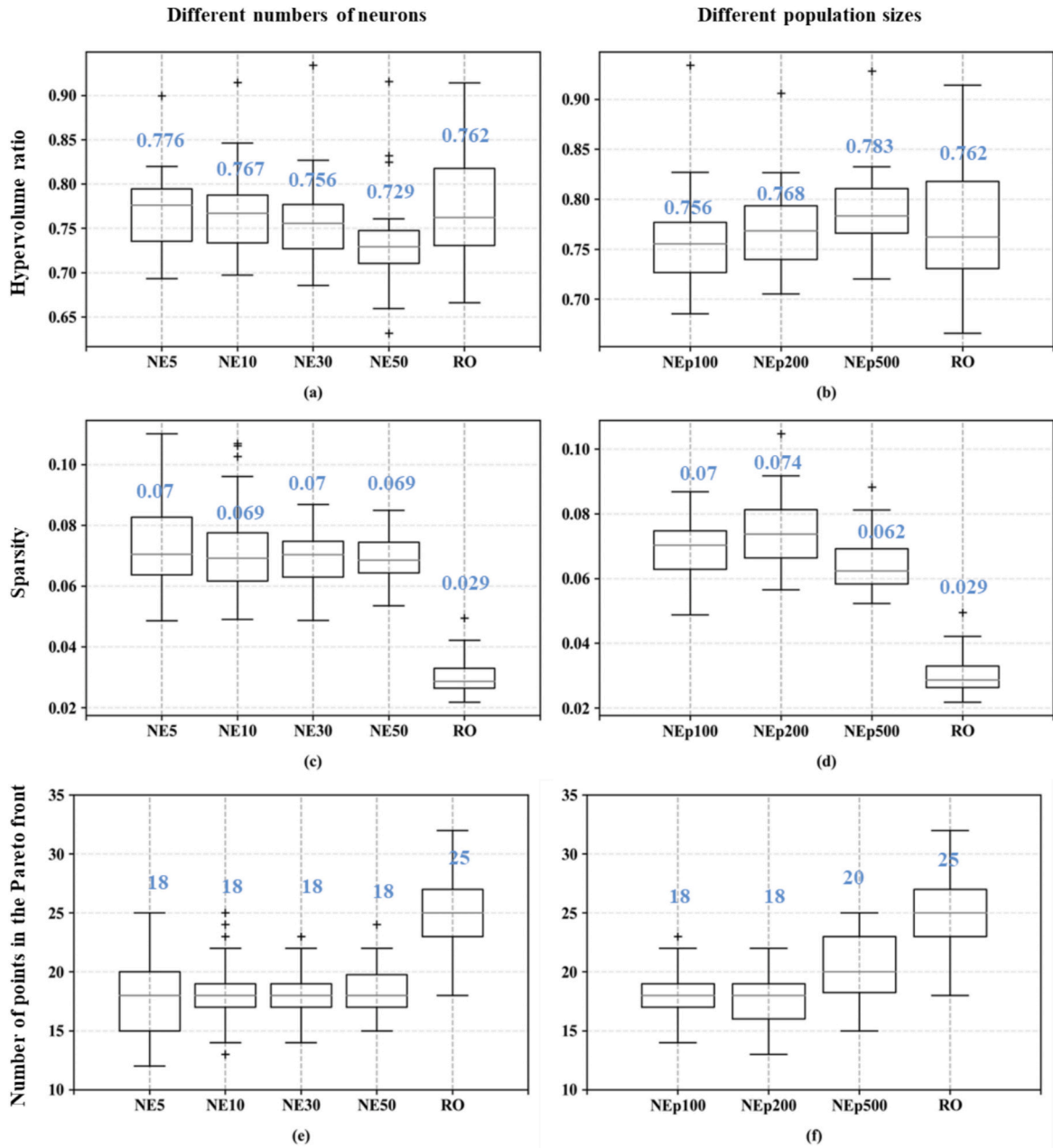


Fig. 5. Performance evaluation for Pareto front approximation.

Table 1

The total number of constraint violations via the neuro-evolution control policies across testing demand scenarios.

|                            |               | NE5 | NE10 | NE30<br>(NEp100) | NE50 | NEp200 | NEp500 |
|----------------------------|---------------|-----|------|------------------|------|--------|--------|
| Number of control policies |               | 100 |      |                  |      | 200    | 500    |
| Constraint                 | Node pressure | 3   | 0    | 1                | 1    | 0      | 19     |
|                            | Pump switch   | 6   | 10   | 7                | 11   | 9      | 3      |

A total of 66 pipe failure events in the default water demand scenario are used to evaluate the performance of the non-dominated pump solutions or control policies in the approximated Pareto fronts. NEp200 is selected here as an illustrative example of neuro-evolution to compare with SSO and RO, as its number of non-dominated control policies matches the number of solutions for SSO and RO (i.e., 200). Each non-dominated pump solution or control policy undergoes evaluation

across all pipe failure events.

The results of SSO are shown in Fig. 8(a), in which the legend related to resilience is measured by mean MRI under normal conditions of the default water demand scenario. The solutions with high resilience demonstrate resistance to negative impacts caused by pipe failure and closure, whereas those with low resilience are likely to cause pressure deficient events, thereby decreasing the water supply ratio. In Fig. 8(a),

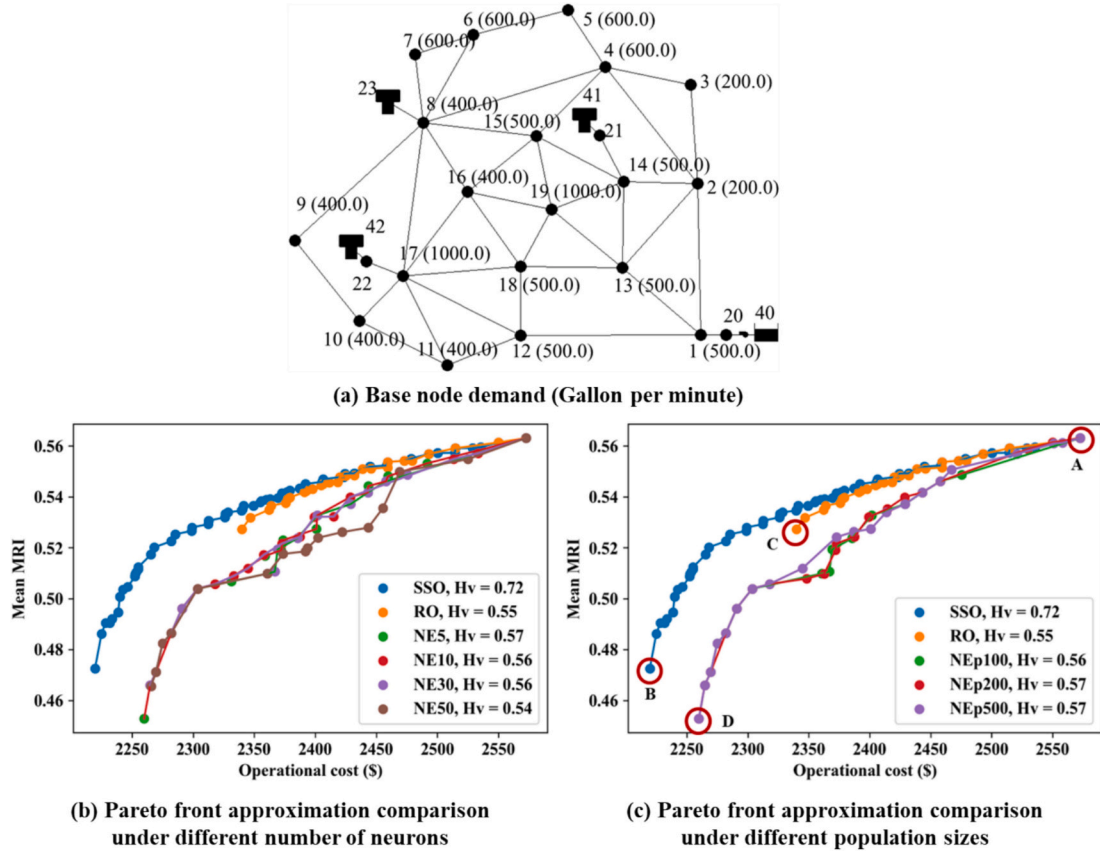


Fig. 6. Performance of Pareto front approximation under the default water demand scenario.

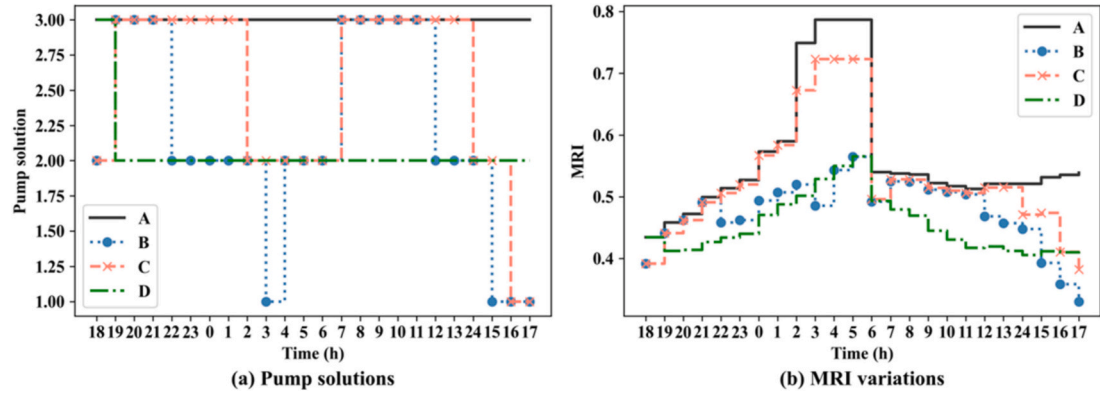


Fig. 7. Pump solutions and MRI variations of the Solution A, B, C and D.

17 % of pump solutions in the Pareto front lead to pressure deficient events. Particularly, the pump solution of SSO with the lowest resilience results in 59 pressure deficient events out of 66 pipe failures, with a minimum RWS of 93 %. The results of RO in Fig. 8(b) indicate that all non-dominated pump solutions, which exhibit relatively high resilience, achieve a 100 % water supply ratio in all pipe failure events. There are no pump-switch violations (i.e.,  $NV = 0$ ) for both SSO and RO. The results of NEp200 are presented in Fig. 8(c). NEp200 has only one control policy causing one pressure deficient event and three control policies violating the pump-switch constraint. Unlike SSO, NEp200's control policies with low resilience still show commendable performance in pressure management and water supply. From a water utility perspective, neuro-evolution and RO outperform SSO under pipe failure events. This may be attributed to the fact that SSO solutions are fixed after

optimization for each specific scenario. When pipe bursts alter the hydraulic state of the network, these static solutions may become less effective. In contrast, neuro-evolution policies are adaptive and can dynamically adjust their outputs (i.e., pump solutions) in real-time based on changes in water levels, thus preventing water pressure deficiencies. Thus, even the least-operational-cost (i.e., the least-resilience) control policy of neuro-evolution is still effective in real-time control under pipe failure events.

Further, the least-operational-cost solution of RO and different least-operational-cost control policies of neuro-evolution are applied to all testing scenarios (i.e. a total of 3300 pipe failure events) for a comprehensive performance comparison. In Fig. 9(a), except for NE50 that leads to 49 pressure deficient events, neuro-evolution control policies generally have fewer pressure deficient events than RO. NE50, however,

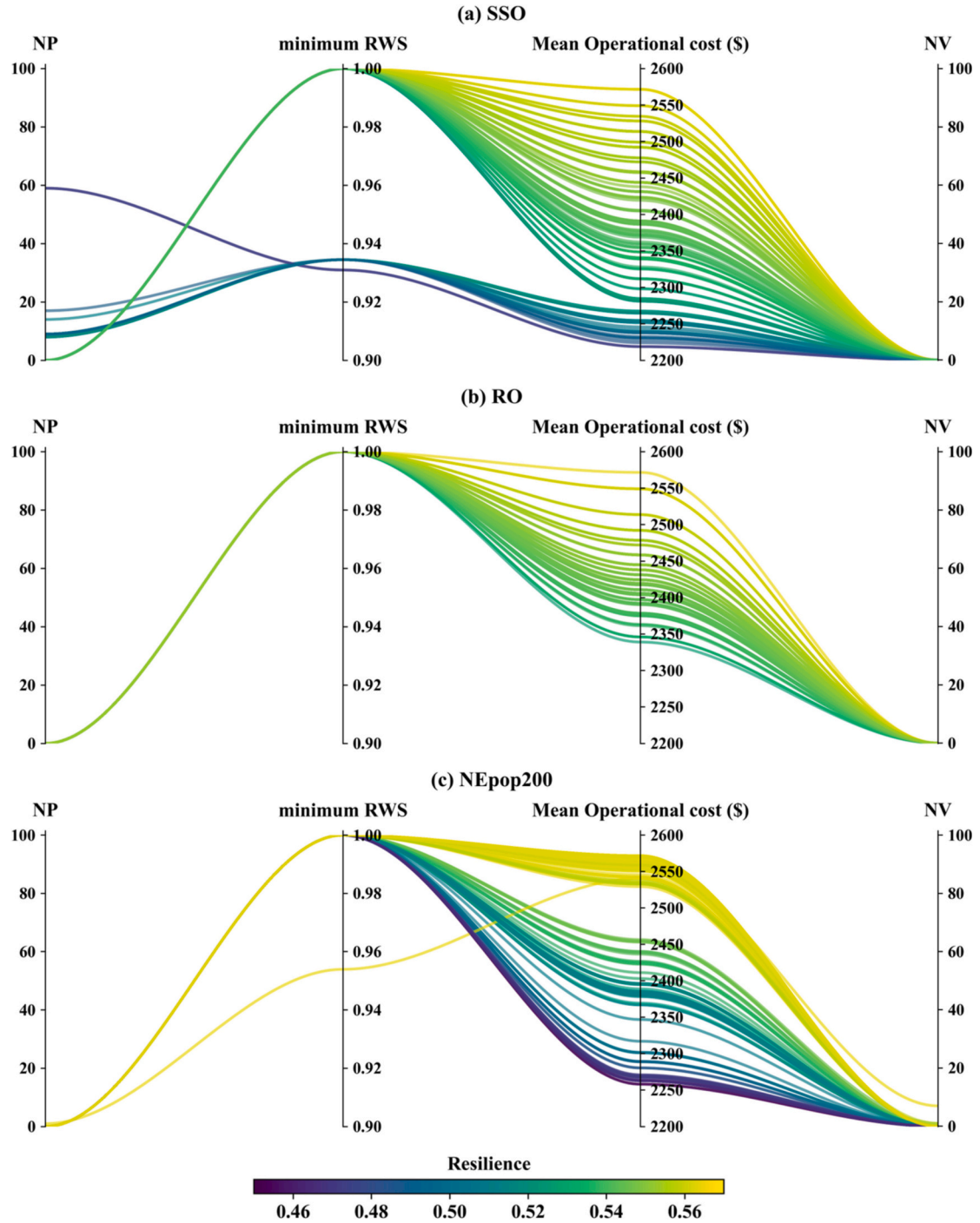


Fig. 8. Performance evaluation under pipe failure events of the default water demand scenario.

perform better in water supply compared to RO. Fig. 9(b) indicates that the minimum RWS in the pressure deficient events of all neuro-evolution control policies is very close to 100 %, whereas that of RO is just about 92 %. Moreover, the mean operational cost of neuro-evolution control policies is at least 2.7 % less than RO (see Fig. 9(c)). Despite a few pump-switch constraint violation, as depicted in Fig. 9(d) with a maximum NV is just 20 for all pipe failure events, neuro-evolution demonstrates superior cost-effectiveness over RO under pipe failure events, considering the ratio of water supply and operational cost.

## 5. Conclusions

This study presents a real-time multi-objective optimization method for pump scheduling in WDNs based on neuro-evolution. A resilience index (i.e., MRI) and operational cost are incorporated into the optimization model simultaneously. Based on the simulation results of the Anytown network, the main conclusions are drawn as follows:

- (1) The performance of the neuro-evolution falls between SSO and RO concerning the Pareto front hypervolume metric. The utilization of a smaller-scale neural network and a larger population size in the NSGA-II algorithm contribute to improve the

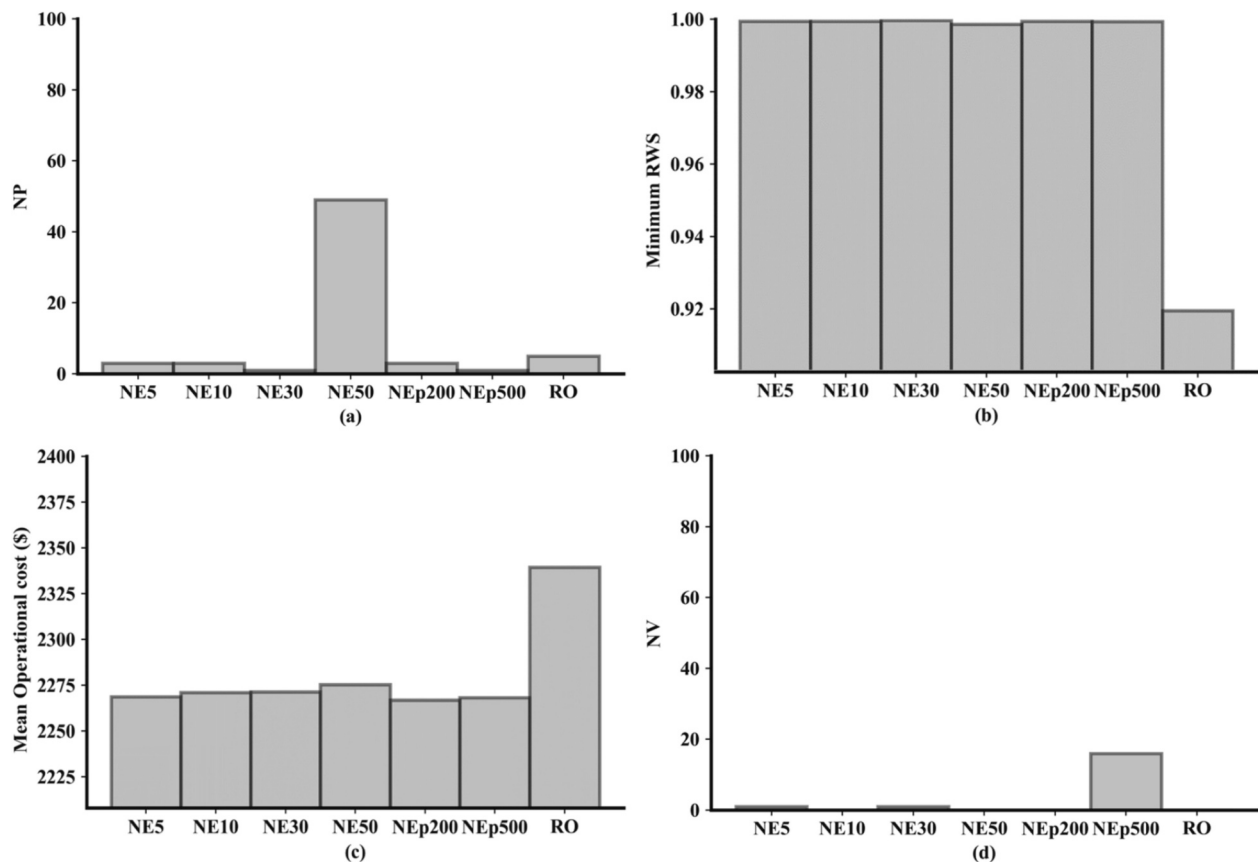


Fig. 9. The least-operational-cost solution performance of neuro-evolution and RO under various pipe failure events.

hypervolume of the approximated Pareto front. However, neuro-evolution exhibits inferior performance in the approximated Pareto fronts sparsity, when comparing to RO. Further exploration is needed to enhance the quality of the approximated Pareto front through neuro-evolution.

- (2) While the approximated Pareto front of SSO almost entirely dominates those of neuro-evolution in the default demand scenario, the advantages of SSO are more significant in the low resilience area compared to high resilience area. In addition, the approximated Pareto front of neuro-evolution exhibit broader resilience ranges than RO, as RO struggles to provide effective pump solutions in the low resilience area.
- (3) Neuro-evolution outperforms SSO and RO under pipe failure events. The simulation of pipe failures in the default water demand scenario shows that over 99 % non-dominated neuro-evolution control policies in the Pareto front effectively prevent water pressure deficiencies, whereas only 83 % of non-dominated solutions obtained through SSO address this issue. The simulation of pipe failures across all testing scenarios shows that the neuro-evolution control policy with the least operational cost shows a 2.7 % reduction in mean operational cost compared to RO under the pipe failures across all testing scenarios, while also achieving approximately an 8 % higher minimum water supply ratio. Neuro-evolution is effective for pump operation under pipe failures.

#### CRedit authorship contribution statement

**Shengwei Pei:** Writing – original draft, Methodology, Investigation, Formal analysis, Conceptualization. **Lan Hoang:** Writing – review & editing, Supervision. **Guangtao Fu:** Writing – review & editing, Visualization. **David Butler:** Writing – review & editing.

#### Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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#### Data availability

The algorithm examples used in this study are available at [https://github.com/githubwateruse/MORTC\\_WDN](https://github.com/githubwateruse/MORTC_WDN).

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